INCH-POUND

MIL-S-80224A
19April 1989
SUPERSEDING
MIL-S-80224
15 July 1971

MILITARY SPECIFICATION

SHEARING MACHINE, METAL NOTCHING, POWER DRIVEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of power driven metal shearing and notching machines for cutting 900 corner and vee shaped notches in sheet and steel plate, ccmplete with equipment and accessories specified herein.

1.2 Classification. The shearing machine shall be of the following types. The size shall be determined by material thickness and corner notching capacity. The type and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Single station - 900 corner and vee notching, variable angle cutting and chamfering operations.

Type II - Double station - 900 corner and vee notching, variable angle cutting and chamfering, punching, edge notching, and corner radius trimming operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneflclal comments (recommendations, add ltlons, deletlons) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd. Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC NIA FSC 3445

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contract activity.)

"

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

2.2 Non-Government @ubl ications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which" are DOD adopted are those listed in-the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the,documents c!ted in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 218.01 - Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth.

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for 8evel, Hypoid, Fine Pitch Wormgearing and Racks only as Unassembled Gears.

AGMA 2000-A88 - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, Suite 201, 1500 King Street, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B1l.4 - Shears - Safety Requirements for the Construction, Care, and Use. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363) .

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Tool Builders Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

NAT ONAL FIRE PROTECTION ASSOC ATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD Adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and t~c=rences cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Design. The machine shall be a self-contained power driven unit designed to perform a variety of sheet metal shearing operations to incl ude 900 corner notching, vee notching, punching, edge notching, chamfering, radius trimming and variable angle cutting. The machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be readily accessible for adjustment, replacement, or repair.

3.2.1 Measuring and indicating device calibration. Measuring and indicating devices shall be graduated in inch-pound (see 6.4.1), metric (see 6.4.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2) . Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate die clearance or blade movement shall be g-ted in increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.1.2 D~ital readout system. When specified (see 6.2), a digital readout syslem shall be installed on the X and Y axes of the workRiece squaring guide. "rhe system shall consist of a position feedback device; an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shallprovide a re~dily visible dig' tal display over the full working range of each axis on which installed. The system resolution shall be 0.005 inch. The system accuracy shall be equa" to the system

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.005 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch/pound and metric units. An input power source fluctuation of:10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.2.1.2.2 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch-metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work station. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The machine and its components that directly consume energy In normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station.

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assem~=s furnished on the machine, whether or not specifically required herein, shall conform with all requirements of 29 CFR 1910 and ANSI B11.4. If a conflict arises between 29 CFR 1910 and B1l.4, 29 CFR 1910 shall apply. Additional ia[cl:y and health requirements shall be as specified in detail (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine th,lt present safety hazards. Additional guarding dependent on workpiecc size ~nd configuration shall be provided when specified and fully describe'd hy t.hc procuring activity (see 6.2).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Ashcstos restriction. Asbestos and materials containing asbestos shall not=sed on or In he machine.

3.2.8 Environmental protection. During operation, service, transportation, or stor~ge of the machine, it shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation.

3.2.9 Lubrication. Means shall be provided to ensure complete lubrication for all ~-rts. When specified (see 6.2), a centralized manual lubrication system with positive adjustable lubricant delivery rate, automatic high pressure relief mechanism, and cleanable or replaceable in-line filter shall be provided. The lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

3.2.10 Interchan eability. To provide for replacement of worn parts, all parts ~hall be ma'hu+ actured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, 'c?t~i, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. A" 1 fasteners shall be installed to prevent change of Gijhtness. Fastening devices subject to removal or adjustment shall not be permanently installed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded *or* beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Gears. All gears of the ram and drive train shall be designed and manufactured from a suitable ferrous material to meet or exceed the requirements of AGMA 218.01, AGMA 390.03a, and AGMA 2000-A88 for the inch-pound system or ANSI/ 1S0 54 for the metric (S1) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

The power driven metal notching machine shall consist of n;''le=~ frame work table ram, drive system (mechanical or hydraulic), electrical s~stem, and ot~er components required to meet the requirements specified herein.

3.4.1 Frame. The frame shall support the work table, ram, machine drive, and all m components specified herein. The frame structure shall be either cast iron, cast steel, or a fabricated steel weldment (stress relieved) of the mass, weight, and rigidity to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The frame shall be a floor mounting unit, arranged to support the work table at a working height of not less than 34 inches, and not more than 36 inches from the mounting height. The base of the frame shall have means to permit leveling and fastening the machine to the floor or foundation. The frame enclosure shall have covered access openings of the size required to facilitate servicing, maintenance, and adjustment of all components contained therein. The control panel at the front of the machine shall contain all electrical controls for manual or automatic operation. When specified (see 6.2.), an optional turn table with locking mechanism shall be provided in the base of the machine. The turntable shall allow the machine to be positioned 900 and 1800 from the original work position, conserving work space and permitting floor space flexibility. The frame shall incorporate a chute for discharge of off-all or scrap. The chute shall be sufficient in size to

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

handle the maximum capacity part of the machine. An eye or an attachment point shall be provided at or near the machine's balance point for lifting the machine. The eye or lift point shall be capable of supporting five times the weight of the machine, and its location shall not cause permanent distortion when suspended.

3 4.2 Worktable. The worktable shall be a separate component supported by and attacfied to he frame. The worktable shall allow for easy positioning of the workpiece under the ram for notching operations. The squaring guides and bui" t in scales shall align the workpiece in position for the variety of dif. went notches that can be made. The worktable shall be either one piece cast iron or cast steel of the mass, weight, and rigidity to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The top of the worktable shall be machine finished to 125 micro inches or less, and shall be flat to within a tolerance of +0.005 inch per foot. The worktable shall incorporate T-slots to accept squ~ring guides. The worktable shall include lower blades secured in position for notching. The blades shall be reversible of high carbon, high chrome alloy steel, hardened to not less than 57 nor more than 62 Rockwell C, and machine finished for endurance and accuracy. The worktable of the type II machine shall provide T-slots at the rear of the table for the rear punching station.

3.4.3 Ram. The ram shall be either cast iron, cast steel, or a fabricated steel weldiiii%t (stress relieved) of the mass, weight, and rigidity to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The ram is designed for supporting and driving the upper notching blades. The design shall be such that the upper blades shall be reversible and choice of cutting methods shall be point to rear or rear to point. The ram for the type II machine shall conform to the requirements of table II to provide a punching station at the rear of the machine opposite the vee notching station. The ram shall provide means for adjustable, spring loaded, hold-down devices to ensure that the material being cut does not move during the notching operation. The ram's reciprocating motion shall be accomplished on machined bearing surfaces, hardened, and machine finished. The ram shall contain sealed bearings or grease fittings implemented to ensure prolonged service. A guard shall be installed around the ram to prevent accidental access to the blade area when the machine is in operation.

3.4.4 Drive. Unless otherwise specified, the notching and punching action of the mac=shall be accomplished by either a mechanical or hydraulic type drive. When only one drive is acceptable, the particular type drive required shall be as specified (see 6.2). The drive shall be designed to assure that sufficient energy is developed and supplied to the ram for accomplishing the performance requirements specified herein. The machine shall provide means for adjusting the ram stroke without stopping the machine. The standard operating modes of the machine shall be single cycle and continuous cutting. When specified (see 6.2), optional modes of operation shall be provided such as inching and time delay. The machine shall include provisions for only one stroke of the ram for each depression of the foot pedal drive control.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

TABLE 1. Type I, Notching Machine, Sizes, and Capacities.

Characteristics	Capacities		
	Size 1	Size 2	Size 3
Maximum material, thickness	5/32	1/4	318
Maximum corner notch	8 1/2 x 8 1/2	8 l/2x 8 112	11 314X 11 314
Drive motor - HP	4.0	5.5	10.0
Dial adjustment die clearance	No	Yes	Yes

TABLE II. ~~Type II, Notching~~ and Punching Machine, Sizes, and Capacities.

Characteristics	Capacities			
	Size 1	Size 2	Size 3	Size 4
Maximum material, thickness	118	5/32	7/32	1/4
Maximum corner notch	a 112X 8 1/2	8 1/2 x 8 1/2	8 5/8 x 8 5/8	10 x 10
Punching capacity, tons	11	8	27	12
Drive motor - HP	3.0	5.0	5.0	10.0
Dial adjustment die clearance	Yes	No	Yes	Yes

NOTES: Machine size and capacity shall be not less than the stated ~~- quirements of the size ordered nor greater than the requirement in the the next larger size shown, excluding $P^{unchin}gs_{ize}$ capacity, and capacity above tables. Offered units may exceed the requirements of the above tables by no more than 10%. Machines exceeding these requirements may be considered~ but it shall be at the discretion of the procuring activity to determine acceptability of non-complying offers.

All dimensions in the above tables are in inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

3.4.4.1 Mechanical type drive system. The mechanical type drive system shall include an electrlc motor driven reduction aearbox to transfer force to the punching ram. The drive shall convert rot~ry motion to reciprocating motion. The drive shall consist of an electric motor conforming to 3.4.5.1 and a belt drive arrangement powering a driven eccentric system. Rotating components shall be supported on bearing of sizes, types, and precision classes required for the specific applications to ensure maximum service life and accuracy. The drive shall be enclosed in and supported by the machine frame.

3.4.4.2 Hydraulic type drive system. The hydraulic type drive system shall confor"m to or exceed the requirements of JIC H-1 and shall be com~lete, including all pumps, valves, piping, cylinders, and pressure controls." Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have a means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable. The ram cylinder unit shall be provided with overload protection by a pressure relief mechanism or pressure limit switch on the ram.

3.4.5 Electrical requirements. The machine shall be wired to conform to NFPA 79. Unless otherw~se specified (see 6.2), each machine shall be designed to operate on 3-phase, 60-Hertz (Hz), 230/460-volts. Unless otherwise specified (see 6.2), the machine shall be initially wired for operation on 460 volts.

3.4.5.1 Motors. Motors shall meet the requirements of 3.4.5. Motors shall be rat- continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

3.4.5.2 Solid state components. Solid state design shall be used throughout for electronic components.

3.4.5.3 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.4.5.4 Worklight. When specified (see 6.2), an incandescent worklight shall be provided. The fixture shall have an air vented protective reflector shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. The lamp shall be no less than 75 watts.

3.4.5.5 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 9,999 hours in increments of not greater than one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

3.4.6 Die Clearance. Die clearance shall be preset to accommodate most material types' and thickness. Means shall be provided to allow additional adjustment to the dies when required. When specified (see 6.2), a die clearance adjustment dial shall be provided to ensure the most suitable clearance for a specific material thickness and type, providing burr-free cutting.

3 . 5 ~ . Unless otherwise specified (see 6.2), the size and capacity of the machine shall be not less than those specified in table I or table II for the type and size machine specified.

3.6 Performance. The machine shall be capable of performing sheet and steel plate shearing operations in the workpiece thicknesses specified in table I or table II for the type and size machine required. The machine shall be capable of performing the metal working operations in workplaces of mild steel having not less than 50,000 PSI tensile strength without deforming the workpiece.

3.6.1 Type I machine. The type I machine shall be capable of performing sheet and steel plate corner and vee notching, variable angle cutting, and chamfering when using the squaring guides, scales, and gauges. The machine shall be capable of performing shearing cuts in shim stock up to the maximum material thickness and corner notching capacity.

3.6.2 Type 11 machine. The type II machine shall be capable of performing all of the sheet and steel plate operations of the type I machine as well as punching, edge notching, and corner radius trimming operations. The machine shall be capable of performing shearing cuts and punching operations up to the maximum material thickness and corner notching capacity.

3.6.3 Additional performance requirements. Performance features required in addition to, or In lieu of, he requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally furnished with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2).

3.9 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.10 Marking *on* instruments, control panels, charts, and plates. All words on instruments. control oanels. charts. and plates shall be in the English language. Characters shal'1 be permanently marked in boldface on a contracting background. All plates shall be corrosion resistant,

3.10.1 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Points of lubricant application
Type of lubricant

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

Servicing interval
Viscosity
Military or Federal specification for each lubricant
(if applicable)

3.10.2 Nameplate. A corrosion resistant nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National stock number or plant equipment code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When specified (see 6.2), a corrosion resistance hydraulic fluid requirements plate shall be attached to the machine at the hydraulic reservoir fill point. Unless otherwise specified (see 6.2), the plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the machine and its accessories shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, he contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program, The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the first article examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.3. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the machine is installed at its final location. The acceptance test shall consist of the examination *in* 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 410 to 1040 Fahrenheit (F) and 20 percent to 95 percent relative humidity.

4.8.2 Operational test. Each machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

4.8.3 Performance tests. The type I machine's performance shall be tested by operating the machine to perform corner and vee notching in the maximum workpiece thickness and size cut to determine conformance with the requirements of 3.6 and table I. The machine shall shear not less than six workplaces of the maximum size and thickness as specified in table I. The type II machine's performance shall meet or exceed the requirements of the type I machine. Additionally, the type II machine's performance shall be tested by operating the machine to perform punching and edge notching in the maximum workpiece thickness to determine conformance with the requirements of 3.6 and table II. The machine shall punch and edge notch not less than six workplaces of the maximum size and thickness as specified in table II.

4.8.4 Additional performance tests. Performance tests required in addition to or in lieu of the requirements of 4.8.3 shall be as specified and fully described (see 6.2).

4.8.5 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requiremmts. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The machines covered by this specification are intended to incorporate simple design, easy set up, and rapid ad.iustment for use in short run sheet metal shearjng, no~ching, chamfering and bunching operations. A variety of operations may be performed on the same machine without expensive tooling and movement of job to other machines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the s~ecific issue of individual documents referenced (see 2.1.1 and 2.2).

d. ~irst article, if required (see 3.1).

e. Specify system of units for measuring and indicating devices (see 3.2.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

f. Digital readout system, if required (see 3.2.1.2).

g. Specify any special digital readout system features, if required (see 3.2.1.2).

h. Additional safety and health requirements, if required (see 3.2.5).

i. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

j. Centralized manual lubrication system, if required (see 3.2.9).

k. Painting, if different (see 3.3.5).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

m. Turn table with locking mechanism in base, if required [see 3.4.1).

n. If machine drive is to be a specific type, state required type (see 3.4.4).

o. Specify and fully describe additional modes of operation, if required (see 3.4.4).

p. Electrical system, if different (see 3.4.5).

q. Initial input voltage, if different (see 3.4.5).

r. Motors, if different (see 3.4.5.1).

s. Worklight, if required (see 3.4.5.4).

t. Hour meter, if required (see 3.4.5.5).

u. Die clearance adjustment dial, if required (see 3.4.6).

v. Size and capacity, if different (see 3.5).

w. Additional performance features, if required (see 3.6.3).

x. Specify and fully describe optional equipment, if required (see 3.8).

y. Specify applicable repair parts, if required (see 3.9).

z* Specify information to be provided on lubrication plate if different (see 3.10.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

aa. Nameplate, if different (see 3.10.2).

bb. Hydraulic fluid requirements plate, if required (see 3.10.3).

cc. Hydraulic fluid requirements plate, if different (see 3.10.3).

dd. Inspection conditions, if different (see 4.3).

ee. First article inspection, if different (see 4.4).

ff. Quality conformance inspection, if different (see 4.5).

gg. Acceptance test, if different (see 4.6).

hh. Additional performance tests, if required (see 4.8.4).

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specification acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.2.2 Technical data. All technical data furnished should be written in the English language.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.8. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions.

6.4.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Bureau of Standards. Inch-pound units having the same names in other countries may differ in magnitude.

6.4.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International d'Unites (S1)," of the International Bureau of Weights and Measures. These units are described in ASTME 380 and IEEE 268 (MIL-STD-961).

6.5 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.6 Cross-reference of classification. The machine classification changes from the previous issue of this specification reflects changes in capacity and performance requirements. New sizes were added for type I and type II machines.

MIL-S-80224	MIL-S-80224A
TYPE I	TYPE I
Sizes (None)	Size 1 (New size)
	Size 2 (New size)
	Size 3 (New size)
TYPE II	TYPE II
Size (None)	Size 1 (New size)
	Size 2 (New size)
	Size 3 (New size)
	Size 4 (New size)

6.7 Changes from previous issues. Marginal notations are not used in this revision to ldentlfy changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. The 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

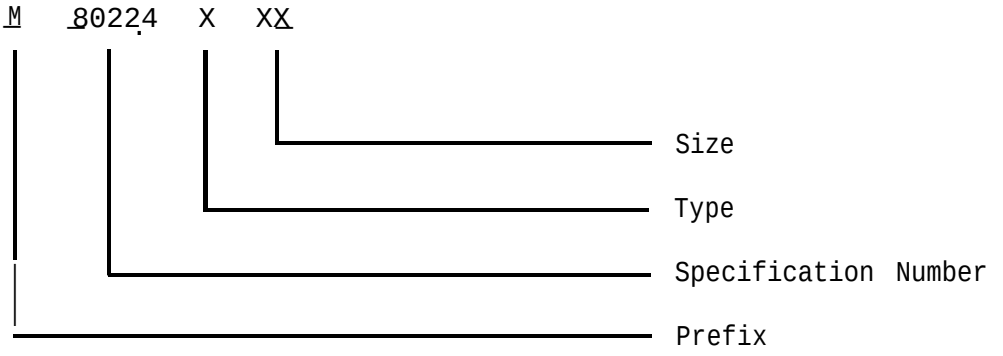
6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

6.10 Warrant Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for notching machines acquired to this specification are created as follows:



6.13 Subject term (keyword) listing.

Chamfering
Notching
Punching
Shearing

Custodians:
Air Force - 99

Preparing activity:
DLA - 1P

Project (3445-0114)

Review activities:
Navy - SH, OS
Air Force - 84

User activities:
Navy - SH, OS

d_I.K GOVFXNMW IWNTING OFFICE IW9 - 604433/0032

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In s continuing ● ffort to maka our standardization documenti better, the DoD pi-idea Ma **form for uee in submitting comrnen~ and mggeationa for improv=menk.** AN USMC **of** military standardization dccumenta are intited to provide Nggationa. 'IMu form nmy **b.** detached, folded along the tines indicst-L taped along the l-e edue (DO NOT STAPLE), **and tiled. In** block s, be u specific M Poeaible about p-icular problem U- -uch M wording Which =wi~d in~wret~~ion, - too rigid, rwtrictiw, 1oo-, am%iguow or waa incompatible, and give prowaed wording chanw which would alleviate the **problem. Enter in block 6 ● ny rernarka not relded to a specific** paragnph of the **document. If block 7 u filIed** out, an ● **cknowledgement will be mailed to you within 30 AYG to let you** know **tbst** your comments were **=eived** md are **being** csmeidered.

NOTE: Tbie form may not be UA to request copiaa of documents, nor ta request waivers, deviation.e, or clarification of ~ification requirement on **curnnt contmcti.** Conment.z **submitted on** this Corm do not conatitnta or imply authori=tion to waive any portion **of the referenced document(s) or to amend contmctti requirements.**

(Fold o10811 lhbr the)

(Fold aknv lhb llna)

DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
d\TTx : DIPEC-SSG
}lemphis, TN 38114-5051

1 1 1 1 1 1 F I

ufvl TEO STATES

[redacted]

Or FICIAL EUSINESS
PENALTY FOF4 PRIVATE USE S300

BUSINESS REPLY MAIL]
FIRST CLASS PEAMIT NO. 4966 WASHINGTON D. C

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Defense Industrial Plant Equipment Center
ATTX : DIPEC-SSG
Memphis, TN 38114-5051

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.

srArJoARol"u71cm ticumEnt iIdpROVEMENT propOsal

(See Instructions -Reverse SideJ .

OOCUMENT NUMEJER	2. OOCUM6NT TITLE
iIL-s-80224A	Shearing Machine, Metal Notching, Power Driven

a, NAME 0FsU8M1171N0 ORGANIZAT10N

. AOORESS (9hwt, *CitY, StaU.* ZIP C-)

4. TYPEOF ORGANIZATION (Mor&omc/

- ❑ VENOOR
- ❑ ".SER
- ❑ MANUFACTURE
- ❑ " OTHER (Spec(fi,:

. PRoELEMAIIEAS

a P~rqyfnph Numow ● nd Wording:

b. flocommendti Wo#dlnv:

c. flcnon/Rationalo for R.comwwrtdottart:

;. REYAA$IKS

kNAMEOFSUBMITTEft *(LMf.* Firtf. AffJ - Op~lOnd	*b.* WOYtK TELEPHoN6 NUM8EFY (InciUd8AIT。 Co&) - OptIOn*I
. MAILING AOOf+ESS (S(rWe I, CJf Y, .$ fofu. ZJPCOAI - OPt, on. i	@ OATE. OF SUf3MI SS10N fYY.v.v DD)

DD FORM 82 MAR 1425

● REVIOUS EoITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:47Z
Check the source to verify that this is the current version before use.